MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 15
   - Horsing around
   - Wrapping up

Review

Last time

Review

Last time

- Ready set bet

Review

Last time

- Ready set bet
- Expectation values

Outline

1. SoPS Lecture 15
   - Horsing around
   - Wrapping up

Expected horse position

Problem

What's the expected position of each horse after one roll of the dice?

Expected horse position

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What's the expected position of each horse after one roll of the dice?

- sample space: outcomes of rolling two dice

$$\Omega = \{(1,1),(1,2),(1,3),\ldots,(6,6)\}$$

Expected horse position

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- each horse has a different evaluation function

Expected horse position

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$$\Omega = \{(1,1),(1,2),(1,3),\ldots,(6,6)\}$$

- each horse has a different evaluation function
- horse $2/3$:

$$f_{2/3}((j,k)) = \begin{cases} 1, & j+k = 2 \text{ or } 3 \\ 0, & \text{otherwise} \end{cases}$$

Expected horse position

Horse 2/3 expected position

$$f_{2/3}((j,k)) = \begin{cases} 1, & j+k = 2 \text{ or } 3 \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{E}(f_{2/3}) = \frac{f_{2/3}((1,1)) + f_{2/3}((1,2)) + \cdots + f_{2/3}((6,6))}{|\Omega|} = \frac{3}{36}.$$

Expected horse position

Horse 4 expected position

$$f_4((j,k)) = \begin{cases} 1, & j+k=4 \\ & \text{otherwise} \end{cases}$$

$$\mathbb{E}(f_4) = \frac{f_4((1,1)) + f_4((1,2)) + \cdots + f_4((6,6))}{|\Omega|} = \frac{3}{36}.$$

Expected horse position

Horse 5 expected position

$$f_5((j,k)) = \begin{cases} 1, & j+k=5 \\ & \text{otherwise} \end{cases}$$

$$\mathbb{E}(f_5) = \frac{f_5((1,1)) + f_5((1,2)) + \cdots + f_5((6,6))}{|\Omega|} = \frac{4}{36}.$$

Expected horse position

Horse 6 expected position

$$f_6((j,k)) = \begin{cases} 1, & j+k=6 \\ & \text{otherwise} \end{cases}$$

$$\mathbb{E}(f_6) = \frac{f_6((1,1)) + f_6((1,2)) + \cdots + f_6((6,6))}{|\Omega|} = \frac{5}{36}.$$

Expected horse position

Horse 7 expected position

$$f_7((j,k)) = \begin{cases} 1, & j+k=7 \\ & \text{otherwise} \end{cases}$$

$$\mathbb{E}(f_7) = \frac{f_7((1,1)) + f_7((1,2)) + \cdots + f_7((6,6))}{|\Omega|} = \frac{6}{36}.$$

Expected horse position

Problem

What's the expected position of each horse after one roll of the dice?

Solution:

horse	2/3	4	5	6	7	8	9	10	11/12
exp. pos	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{12}$

Expected horse position

Problem

What's the expected position of each horse after two rolls of the dice?

Expected horse position

Problem

What's the expected position of each horse after two rolls of the dice?

- sample space: outcomes of rolling two dice

$$\Omega = \{(1,1,1,1),(1,1,1,2),(1,1,1,3),\ldots,(6,6,6,6)\}$$

Expected horse position

Problem

What's the expected position of each horse after two rolls of the dice?

- sample space: outcomes of rolling two dice

$$\Omega = \{(1,1,1,1),(1,1,1,2),(1,1,1,3),\ldots,(6,6,6,6)\}$$

- size of sample space:

$$|\Omega| = 6^4 = 1296$$

Expected horse position

Problem

What's the expected position of each horse after two rolls of the dice?

- sample space: outcomes of rolling two dice

$$\Omega = \{(1,1,1,1),(1,1,1,2),(1,1,1,3),\ldots,(6,6,6,6)\}$$

- size of sample space:

$$|\Omega| = 6^4 = 1296$$

sample space getting bigger . . .

Non-uniform probabilty

Non-uniform probabilty

- alternative sample space:

$$\Omega = \{(2/3, 2/3), (2/3, 4), (2/3, 5) \ldots, (11/12, 10), (11/12, 11/12)$$

Non-uniform probabilty

- alternative sample space:

$$\Omega = \{(2/3, 2/3), (2/3, 4), (2/3, 5) \ldots, (11/12, 10), (11/12, 11/12)$$

- size of sample space:

$$|\Omega| = 9^2 = 81$$

Non-uniform probabilty

- alternative sample space:

$$\Omega = \{(2/3, 2/3), (2/3, 4), (2/3, 5) \ldots, (11/12, 10), (11/12, 11/12)$$

- size of sample space:

$$|\Omega| = 9^2 = 81$$

- tradeoff is probabilities are no longer uniform!

Non-uniform probabilty

Problem

What is the probabilty of $(2/3, 2/3)$?

Non-uniform probabilty

Problem

What is the probabilty of $(2/3, 2/3)$?

Theorem

If A and B are independent events, then

$$\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\mathbb{P}(B).$$

Non-uniform probabilty

Problem

What is the probabilty of $(2/3, 2/3)$?

Theorem

If A and B are independent events, then

$$\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\mathbb{P}(B).$$

$$\mathbb{P}((2/3, 2/3)) = \mathbb{P}(2/3)\mathbb{P}(2/3) = \frac{3}{36} \cdot \frac{3}{36} = \frac{1}{144}.$$

Non-uniform probabilty

Problem

What is the probabilty of $(4, 7)$?

Non-uniform probabilty

Problem

What is the probabilty of $(4, 7)$?

$$\mathbb{P}((4,7)) = \mathbb{P}(4)\mathbb{P}(7) = \frac{3}{36} \cdot \frac{6}{36} = \frac{1}{72}.$$

Probability table

probabilities (divided by 1296)

$\mathbb{P}$	2/3	4	5	6	7	8	9	10	11/12
2/3	9	9	12	15	18	15	12	9	9
4	9	9	12	15	18	15	12	9	9
5	12	12	16	20	24	20	16	12	12
6	15	15	20	25	30	25	20	15	15
7	18	18	24	30	36	30	24	18	18
8	15	15	20	25	30	25	20	15	15
9	12	12	16	20	24	20	16	12	12
10	9	9	12	15	18	15	12	9	9
11/12	9	9	12	15	18	15	12	9	9

Non-uniform probabilities

Non-uniform probabilities

- sample space

$$\Omega = \{o_1, o_2, \ldots, o_N\}$$

Non-uniform probabilities

- sample space
$$\Omega = \{o_1, o_2, \ldots, o_N\}$$
- probability of o_i is $p_i = \mathbb{P}(o_i)$

Non-uniform probabilities

- sample space
$$\Omega = \{o_1, o_2, \ldots, o_N\}$$

- probability of o_i is $p_i = \mathbb{P}(o_i)$
- probability of event $E = \{o_{j_1}, o_{j_2}, \ldots, o_{j_r}\}$ is
$$\mathbb{P}(E) = p_{j_1} + p_{j_2} + \cdots + p_{j_r}.$$

Non-uniform probabilities

- sample space
$$\Omega = \{o_1, o_2, \ldots, o_N\}$$

- probability of o_i is $p_i = \mathbb{P}(o_i)$
- probability of event $E = \{o_{j_1}, o_{j_2}, \ldots, o_{j_r}\}$ is
$$\mathbb{P}(E) = p_{j_1} + p_{j_2} + \cdots + p_{j_r}.$$

- expectation of $f : \Omega \to \mathbb{R}$
$$\mathbb{E}(f) = p_1 f(o_1) + p_2 f(o_2) + p_3 f(o_3) + \cdots + p_N f(o_N).$$

Expected horse position

Problem

What is the expected position of the 2/3 horse after two rolls?

Expected horse position

Problem

What is the expected position of the 2/3 horse after two rolls?

$$
\begin{aligned}
\mathbb{E}(\text{2/3 pos}) &= 5 \cdot \mathbb{P}((2/3, 2/3)) \quad \text{2 moves + 3 bonus} \\
&\quad + 1 \cdot \mathbb{P}((2/3, 4)) + 1 \cdot \mathbb{P}((2/3, 5)) + \cdots + 1 \cdot \mathbb{P}((2/3, 11/12)) \\
&\quad + 1 \cdot \mathbb{P}((4, 2/3)) + 1 \cdot \mathbb{P}((5, 2/3)) + \cdots + 1 \cdot \mathbb{P}((11/12, 2/3)) \\
&= 5 \cdot \frac{9}{1296} \\
&\quad + \frac{9}{1296} + \frac{12}{1296} + \frac{15}{1296} + \frac{18}{1296} + \frac{15}{1296} + \frac{12}{1296} + \frac{9}{1296} + \frac{9}{1296} \\
&\quad + \frac{9}{1296} + \frac{12}{1296} + \frac{15}{1296} + \frac{18}{1296} + \frac{15}{1296} + \frac{12}{1296} + \frac{9}{1296} + \frac{9}{1296} \\
&= \frac{45 + 54 + 48 + 60 + 36}{1296} = \frac{243}{1296} = 0.1875
\end{aligned}
$$

Is there an easier way?

Expected horse position

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- probability of 2/3 rolled $p = \frac{3}{36}$

Expected horse position

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What is the expected position of the 2/3 horse after two rolls?

- probability of 2/3 rolled $p = \frac{3}{36}$
- probability of 2/3 rolled twice $\mathbb{P}(2/3 \text{ and } 2/3) = p^2$

Expected horse position

Problem

What is the expected position of the 2/3 horse after two rolls?

- probability of 2/3 rolled $p = \frac{3}{36}$
- probability of 2/3 rolled twice $\mathbb{P}(2/3 \text{ and } 2/3) = p^2$
- probability of 2/3 rolled first only
  $\mathbb{P}(2/3 \text{ then not } 2/3) = p(1-p)$

Expected horse position

Problem

What is the expected position of the 2/3 horse after two rolls?

- probability of 2/3 rolled $p = \frac{3}{36}$
- probability of 2/3 rolled twice $\mathbb{P}(2/3 \text{ and } 2/3) = p^2$
- probability of 2/3 rolled first only
  $\mathbb{P}(2/3 \text{ then not } 2/3) = p(1-p)$
- probability of 2/3 rolled second only
  $\mathbb{P}(\text{not } 2/3 \text{ then } 2/3) = (1-p)p$

Expected horse position

Problem

What is the expected position of the 2/3 horse after two rolls?

- probability of 2/3 rolled $p = \frac{3}{36}$
- probability of 2/3 rolled twice $\mathbb{P}(2/3 \text{ and } 2/3) = p^2$
- probability of 2/3 rolled first only
  $\mathbb{P}(2/3 \text{ then not } 2/3) = p(1-p)$
- probability of 2/3 rolled second only
  $\mathbb{P}(\text{not } 2/3 \text{ then } 2/3) = (1-p)p$

$$\mathbb{E}(2/3) = 5p^2 + p(1-p) + (1-p)p = 0.1875$$

Expected horse position

Problem

What is the expected position of each horse after two rolls?

Expected horse position

Problem

What is the expected position of each horse after two rolls?

Solution:

horse	2/3	4	5	6	7	8	9	10	11/12
exp. pos	0.1875	0.1875	0.2469	0.2971	0.3333	0.2971	0.2469	0.1875	0.1875

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Final thoughts

Reflection

Final thoughts

Reflection

- Suppose that there is no bonus for horses anymore. Work out by hand how this changes the expected value of the 2/3 horse after two dice rolls.

Submit your responses by **Tonight!**